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Impact of star and movie buzz on motion picture distribution and box office revenue

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A R T I C L E I N F O

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A B S T R A C T

This study contributes to research on the impact that consumer buzz has on movie distribution and box office success by examining the impact of buzz generated about the individual stars and about the movie itself. The results indicate that movie buzz is instrumental in boosting box office revenue throughout the theatrical release, not just in the later run, as has been suggested in previous studies. Star buzz can enhance box office receipts during the opening week and can contribute to the public's anticipation of the movie pre-release. However, early buzz can have a negative impact on revenue during subsequent weeks if the movie fails to resonate with the audiences. Model simulations reveal that, even for poorly received films, the overall impact of star buzz is positive because the initial revenue boost normally outweighs the later decline. Thus, this study empirically demonstrates the positive impact of star buzz on revenue, which helps shed light on the long-standing debate regarding the importance of star participation in the success of a movie.

1. Introduction and contribution

Buzz marketing, the "amplification of initial marketing efforts by third parties through their passive or active influence" (Thomas, 2004, p. 64), has become a prominent feature in today's information-intensive and consumer-connected marketplace. Word-of-mouth (WOM) is a related concept used to describe buzz marketing fueled by consumer to consumer (C2C) information exchange. These constructs are critical to the movie industry, where 53% of moviegoers base their movie choices on information received from others (Rosen, 2000). Most previous studies have examined the impact of movie buzz on revenue from later runs – that is, after the opening night or opening week (Duan, Gu, & Whinton, 2008; Elberse & Eliashberg, 2003) – but these studies have neglected to examine the impact of buzz on opening revenue. This study offers several contributions. First, it examines the buzz surrounding both the individual stars participating in the movie and the movie itself. Second, it examines the antecedents and consequences of buzz and recognizes the possible endogenous relationships among both types of buzz, box office receipts, and exhibitor decisions about screening intensity. Finally, it estimates the impact of buzz at the opening and during the later run of the film. The remainder of the introduction elaborates on these issues. In particular, the introduction covers (i) the two types of buzz, (ii) the antecedents and consequences of buzz, and (iii) the two time frames used in this study.

1.1. Two types of buzz

1.1.1. Movie buzz

Industry press and academic literature (Duan et al., 2008; Liu, 2006) point out that movies propelled by consumer buzz do well at the box office (e.g., the new *Star Wars* movies generated buzz months before they opened to long lines at premieres). Buzz is also responsible for the success of many so-called sleeper films (e.g., *My Big Fat Greek Wedding*) that become successful later in their life cycles as consumers spread the word about them. Buzz is further listed as a reason for the success of several independent features that gained momentum due to a strong Internet following. *Tunheim Partners*, a communications consulting company, described the online frenzy that made *The Blair Witch Project* one of the most profitable movies of all time.

> "The Internet is swarming with *Blair Witch* fan sites, Web boards, mailing lists, newsgroups, trailer sites, and general excitement about the movie. ... Internet activity started well before the film's release. ... This pre-release hype created the anticipation for the movie's opening that can drive ticket sales" (eStrategy.com, 2000*)*.

Despite an abundance of statements from the consulting community and industry participants, previous studies have not attempted to explicitly measure the amount of online activity prior to a movie's release (with the exception of a small sub-sample of films in Liu, 2006). Therefore, our current understanding of consumer buzz as a driver of opening week box office sales is limited. This shortcoming is a reflection of the fact that buzz has been measured by online user ratings for individual films that become available *after* the movie is

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released. This study expands these earlier frameworks to include pre-release buzz. Because ratings are available prior to the movie opening only for pre-screened films, we turn to a broader Internet search-based measure of a movie's buzz during the pre-release stage.

1.1.2. Star buzz

Buzz is generated not only around movies but also around movie stars ("buzz factor" is a common industry term describing a star's ability to generate consumer interest). Various events in stars' public and private lives generate consumer interest, giving rise to entertainment programs on television such as *Access Hollywood* and *TMZ*. Individual stars can and often do capitalize on consumer interest through various overt and covert PR activities. Industry insiders measure resulting star buzz by the intensity of Internet searches (e.g., using the *Yahoo! Buzz* factor). This study uses this information and expands the definition of movie-related buzz to include both movie- and star-related aspects of buzz. In other words, the study defines buzz as *the consumer excitement, interest and communication around a project or a participating star that is capable of increasing their visibility with both moviegoers and movie industry participants.*

This is not the first study that has attempted to capture consumer fascination with stars when modeling motion picture performance. Various approaches have been used to assess the influence of star power on box office revenue. Previous studies have used stars' historical box office success (Ravid, 1999), visibility and acclaim such as standings in industry-produced power lists (Ainslie, Dreze, & Zufrynden, 2005; Elberse & Eliashberg, 2003; Liu, 2006; Neelamegham & Chintagunta, 1999; Sawhney & Eliashberg, 1996), or Academy Award wins and nominations (Basuroy, Desai, & Talukdar, 2006; Litman, 1983; Ravid, 1999) as measures of star power. Conceptually, these measures are believed to be tied to stars' ability to obtain financing and distribution and to drive opening week revenue; however, the empirical findings have been mixed. This study concentrates on the fact that star power resides in stars' ability to generate consumer excitement and interest and uses traditional measures of star power as antecedents of star buzz.

1.2. Antecedents and consequences of buzz

1.2.1. Antecedents of movie buzz

Conditions necessary for generating movie buzz include consumer awareness and a large pool of potentially interested viewers. Therefore, big-budget movies should be associated with greater buzz than niche offerings (e.g., foreign films). In addition, after the movie opens, one would expect movies with higher recent box office sales to generate more buzz. More interestingly, the characteristics of movie buzz may influence the volume of buzz. In particular, this study focuses on the valence and consistency of movie buzz as possible antecedents of the volume of buzz that is generated while controlling for other salient factors identified in previous studies.

1.2.2. Antecedents of star buzz

Frank and Cook (1995) emphasize stars' ability to create buzz and capture the consumer imagination as primary sources of star power. The buzz surrounding stars can help cut through the clutter, especially in a technologically advanced environment where some stars become known for their sensationalism and "well-knowness" (Boorstin, 1961; Rein, Kotler, & Stoller, 1987; Schickel, 1973). This study takes a similar conceptual view, using the ability to capture consumer interest (operationalized through the intensity of Internet searches) as a measure of star power and exploring previously studied facets of star power (e.g., Academy Awards or bankability) and other potentially relevant dimensions (e.g., sex appeal and recency of success) as its possible antecedents.

1.2.3. Consequences of buzz

Popular culture and the movie industry present overwhelming evidence of the influence of star and movie buzz on box office sales. In *America's Sweethearts*, a 2001 motion picture, the plot involves producers going to great lengths to fabricate dramatic reality in their quest to draw public attention to their stars and projects and to enhance the buzz around them. As with most works of fiction, there is probably some truth to the film's portrayal because it is a widely held belief that buzz sells movies. Although most industry and academic sources agree that movie buzz is a critical driver of box office sales (Duan et al., 2008; Liu, 2006), some prominent studio executives believe that high levels of star buzz could also be harmful. Sumner Redstone, CEO of *Viacom* (the parent company of *Paramount Pictures*), for example, blamed high levels of publicity surrounding Tom Cruise in recent years for compromising his box office power (Marr, 2006).[1] This study examines the impact of these two types of buzz on box office revenue and on exhibitor decisions about screening intensity.

1.3. Two time frames relevant to this study

Previous literature has found that a film's opening and later-run box office receipts may have different drivers (Basuroy et al., 2006; Elberse & Eliashberg, 2003). Therefore, separate models were estimated for the opening week and the later-run periods. One should expect that the two types of buzz play different roles across these two time periods. It is reasonable to expect that, once the movie itself is unveiled, movie buzz will become more important and the importance of star buzz will diminish. The appropriateness of this assumption is addressed later in this paper. Results of different later-run data splits are presented later as well.

2. Conceptual framework

The conceptual framework used in this study is presented in Fig. 1. It reflects the previously mentioned antecedents of star and movie buzz. It also captures the potential consequences of the two types of buzz and their relationship with each other. Earlier studies (Elberse & Eliashberg, 2003; Liu, 2006) considered WOM to be exogenous. However, Duan et al. (2008) modeled WOM as both a precursor to and an outcome of box office revenue. They showed that (i) box office revenue and WOM valence both significantly influence WOM volume and (ii) WOM volume, in turn, leads to higher box office performance.

Pre-release movie and star buzz are obtained from search-based measures that reflect overall excitement and anticipation related to a title or a top star who is involved in the project. Opening week revenue is assumed to be affected by pre-release buzz. Buzz is also assumed to influence distribution decisions. Furthermore, the effects of buzz on demand and distribution decisions could be amplified for movies with high levels of both movie and star buzz due to possible effects of interaction between the two. Similar dynamics are present in later periods. Finally, the conceptual framework captures potential endogeneity between star and movie buzz for two reasons: first, it is hard to talk about a film without bringing up the stars involved in the project, and second, movies promoted by "hot" celebrities could have a higher buzz factor due to star participation. In fact, stars are often contractually obligated to promote their projects (Liu, 2006).

The framework also incorporates the endogenous relationship between buzz and subsequent box office revenue, where star buzz, movie buzz, and their interaction could impact people's desire to see the film. In turn, box office revenue, which is indicative of the number of people who saw the film, fuels movie and star buzz in future periods. Finally, the framework reflects the endogeneity between the number of screens the movie is playing on and box office revenue.

[1] As a result, the 14-year relationship between Paramount Pictures and Cruise/Wagner Productions was severed.

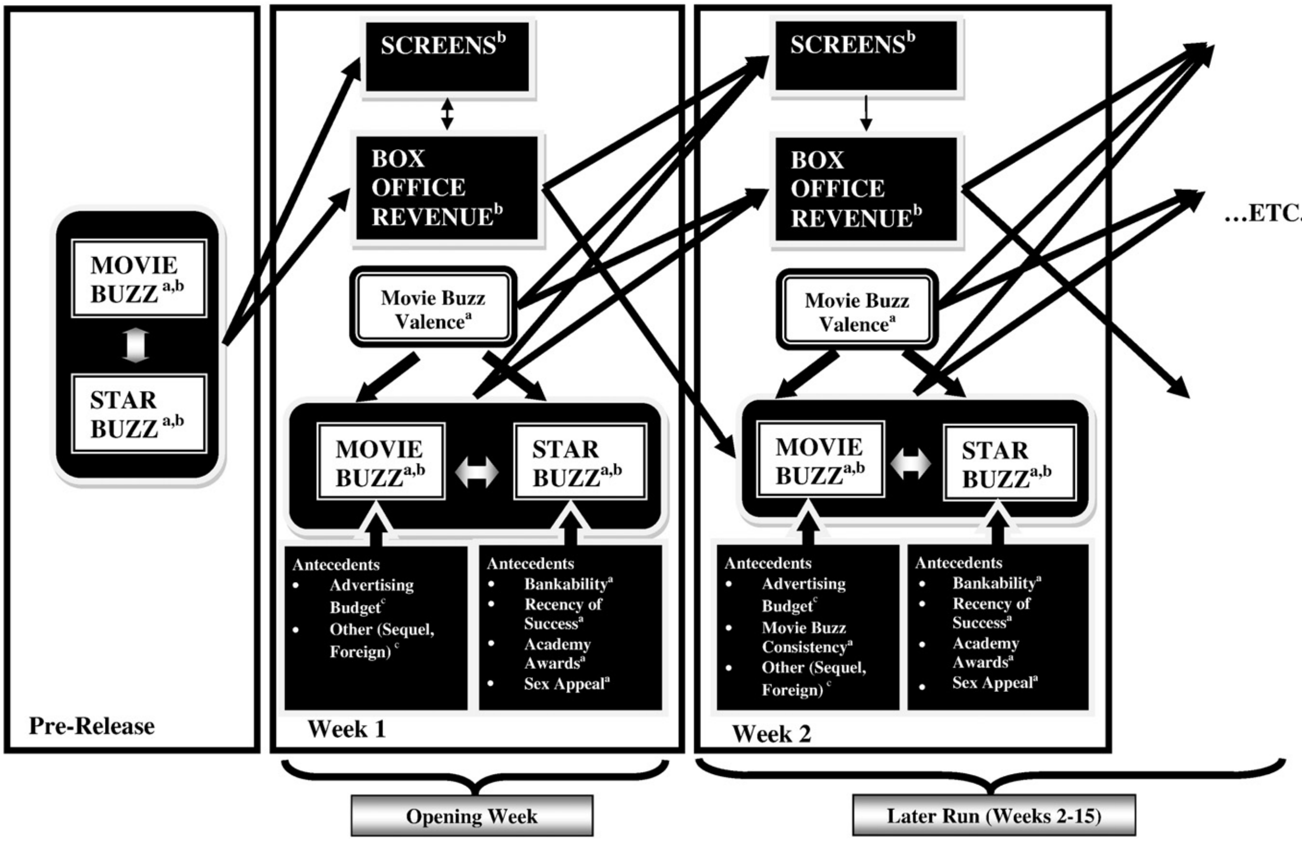

Fig. 1. Conceptual framework. [a]Antecedents of star and movie buzz that are the primary focus of this study. Note that concurrent star buzz fuels movie buzz and vice versa. [b]Factors involved in endogenous supply-demand-buzz relationships in opening week and later-run models. [c]Control variables.

3. Hypotheses

3.1. Hypotheses regarding the antecedents of movie buzz

3.1.1. Valence

Liu (2006) provides the most systematic assessment of the role of consumer buzz in the motion picture industry. He finds that, whereas the volume of WOM is significantly related ($p<0.001$) to revenue, valence is not; in other words, most of the explanatory power of WOM comes from its volume, not its valence. However, the relationship between the valence of buzz and its volume may be more complex; that is, valence could have influence on the volume of WOM (Duan et al., 2008). According to recent studies on movie buzz and consumer expectations (Duan et al., 2008; Wang, Zhang, & Zhang-Foutz, 2007), movie buzz is found to be stronger for well-received movies; if a movie falls short of expectations, movie buzz dies out. These results are consistent with previously extended behavioral explanations suggesting that consumers distribute positive WOM due to altruism/self-enhancement and negative WOM out of vengeance (see Duan et al., 2008, for an extended discussion). Consumers often feel cheated out of their time and money by the poor-quality movies and want to vent their frustration and punish these films by spreading negative WOM. As the valence of WOM goes up, satisfied consumers have greater incentive to spread positive WOM for self-enhancement reasons, whereas dissatisfied customers have greater incentive to spread negative WOM in an attempt to lower the rating in retaliation for bad experience (Duan et al., 2008; Wangenheim & Bayón, 2003). Therefore, one would expect the following:

H1. *The valence of movie buzz is positively associated with the volume of movie buzz.*

3.1.2. Consistency

Following the argument in the previous section, for movies with high overall valence, the few negative statements from dissatisfied customers would increase the variability of ratings and potentially spark a debate regarding the quality of the movie, which would further increase polarization and enhance WOM. In general, movies of all levels of quality desire to be the subject of a debate/conversation that drives people to write reviews. In fact, many forums have a strong debate component, a phenomenon that has led to the emergence of the “argument quality” construct within the electronic commerce literature (Cheung, Lee, & Rabjohn, 2008).

H2. *Movie buzz inconsistency is positively associated with the volume of movie buzz.*

3.2. Hypotheses regarding the antecedents of star buzz

Star buzz is assumed to have several antecedents, as outlined in previous research on star power. This study attempts to distinguish these various drivers of star power and clarify the long-standing debate over the importance of star participation in the success of motion pictures.

3.2.1. Bankability

Rein et al. (1987) argue that paid attendance is the most basic indicator of star power (i.e., past success is a good predictor of future success). Consumers follow the stars who have solid track records and anticipate the future projects of these stars. Successful past projects may serve as conversation starters and lead to expansion of a star's fan base. We will refer to the consistency of a star's past box office success as “bankability.”

H3. *Bankability is positively associated with star buzz.*

3.2.2. Recency of success

Only a few stars remain visible over a long period of time. De Vany and Walls (2005) analyzed 2015 movies released in North America between 1985 and 1996 with a fixed effects model and found that only a handful of stars influence revenue. At the 90% confidence level, only Tom Hanks' or Robin Williams' participation in a film was associated with higher average revenue, whereas Tom Hanks, Michelle Pfeiffer, and Jodie Foster had a positive effect on median revenue. Rein et al. (1987) theorize that stars have life cycles that move through successive stages: emergence, growth, maturity, and decline (albeit with multiple deviations from such paths). These life cycles make recency of success a salient factor for star buzz.

H4. *Recency of success is positively associated with star buzz.*

3.2.3. Academy awards

Although some elitist viewers and industry insiders argue that acting talent and popular appeal or buzz could be orthogonally if not negatively correlated, it is hard to argue that recognition of acting talent creates exposure (e.g., every year, millions of people watch the Academy Awards and discuss the winners, upsets, and latest red carpet fashions). Major awards are capable of creating stars out of relatively obscure actors, and it is a known fact that actor salaries skyrocket following such recognition. It is, therefore, hard to dispute that such recognition fuels consumer interest and communication.

H5. *Academy Awards are positively associated with star buzz.*

3.2.4. Sex appeal

Multiple sources refer to stars as objects of desire and symbols of unattainable beauty. Streitmatter (2004) argues that sex sells in the movies. Attractive celebrities often enjoy a vast following. TV programming is also full of entertainment programs that concentrate on the physical appeal of various stars (50 Hottest Stars of All Time, Greatest Celebrity Bodies, and so on).

H6. *Sex appeal is positively associated with star buzz.*

3.3. Hypotheses regarding the consequences of movie buzz

Previous studies refer to movie buzz simply as "WOM," which is assumed to be exogenous and to play a role only during the later run of the movie (e.g., Elberse & Eliashberg, 2003). Liu (2006) incorporates the fact that buzz can take place before a movie is released, but measures buzz only for a small number of pre-released movies. The current framework is extended to capture people's excitement about films and stars *before* a movie opens. This approach uses Internet search data to measure the amount of buzz surrounding various titles prior to release to ensure that buzz is appropriately captured for *every* movie in the sample. Therefore, the following sub-hypotheses are tested during the opening week and during the later run of the movie:

H7a. *Pre-release movie buzz is positively associated with opening week box office revenue.*

H7b. *Post-release movie buzz is positively associated with later-run box office revenue.*

From a theoretical perspective, valence is an important dimension of Customer Engagement Behaviors (CEB), which are linked to the financial health of the firm (van Doorn et al., 2010). According to the CEB model, "customer actions (e.g., WOM activity … and online reviews) may turn out to be positive or negative for the firm based on the valence of the content" (van Doorn et al., 2010, p. 255). However, the empirical evidence regarding the connection between the valence of WOM and revenue is mixed. Liu (2006) finds no connection between the valence of WOM and box office revenue, whereas other researchers (e.g., Dellarocas & Narayan, 2006; Dellarocas, Zhang, & Awad, 2007; Godes & Mayzlin, 2004) find that the valence of movie buzz is informative when forecasting revenue. When exploring the impact of buzz, we also control for its consistency.

H8. *The valence of movie buzz is positively associated with box office revenue.*

3.4. Hypotheses regarding the consequences of star buzz

The conceptual framework presented earlier in the text models several direct, endogenous, and moderating relationships between star and movie buzz and other variables. Star buzz can be especially critical during opening weekend, when the movie quality is still unknown. As previously mentioned, there are conflicting views about the role of star buzz. The first stipulates that consumer interest unequivocally leads to greater box office success; the second is that star buzz is irrelevant after the movie quality becomes apparent because the movies themselves become the stars (De Vany & Walls, 2005). The last of these views holds that movies should speak for themselves and that buzz around actors can actually be harmful to their projects. As with movie buzz, Internet search data allow us to test the following sub-hypotheses during the opening week and during the later run of the movie:

H9a. *Pre-release star buzz is positively associated with opening week box office revenue.*

H9b. *Post-release star buzz is positively associated with later-run box office revenue.*

H9c. *Star buzz is more important for boosting opening week than later-run revenue.*

3.5. Hypotheses regarding the interaction between the two types of buzz

Star buzz also could interact with movie buzz when "hot" projects with highly visible celebrities garner an unusual volume of audience response. Because this is the first study to examine the effects of star buzz, no previous studies have examined this interaction. This relationship should be especially evident during the later run of the movie, when star buzz could have a positive influence on revenue for well-received movies. For example, people may be more likely to talk about a good movie if it has a more-buzzed-about star. People may also be more likely to remember other viewers' recommendations about such movies. In contrast, if the movie itself fails to captivate audiences, star buzz could backfire and have a dampening effect on revenue. PR hype surrounding the stars could amplify consumer disappointment if the product does not live up to their expectations.

H10a. *Interaction between star buzz and movie buzz is positively associated with opening box office revenue.*

H10b. *Interaction between star buzz and movie buzz is positively associated with later-run box office revenue.*

Star and movie buzz could also influence sales by influencing exhibitor decisions. Conventional logic dictates that exhibitors should allocate more screens to films that are in high demand. However, previous research has found that exhibitors often allocate less screen space to the movies that are expected to have "longer legs" (Elberse & Eliashberg, 2003). In addition, industry literature (Morgan, 2009) states that exhibitors have very little incentive to "push" films during the opening week because it is more profitable to sell tickets during the later run. Therefore, we abstain from formulating a directional hypothesis but acknowledge that star buzz, movie buzz, and their interaction could affect exhibitor decisions.

H11a. *Star buzz influences exhibitor decisions.*

H11b. *Movie buzz influences exhibitor decisions.*

H11c. *Interaction between star and movie buzz influences exhibitor decisions.*

4. Data

This study uses several types of online search-generated data. The Internet Movie Database's (IMDB) search history is used to construct search-related variables. IMDB is a leader among movie portals; it had over 42 million visitors each month during the time these data were collected (the current customer base is 50 million). It is the first sponsored link brought up when a consumer searches for a particular celebrity or a movie title on Google. IMDB allows consumers to search through movie and star archives and traces these activities to capture both movie and star buzz. The site's *STARmeter* and *MOVIEmeter* tools track weekly star and movie rankings based on user search intensity. This study also uses boxofficemojo.com, the-numbers.com, entertainment news archives, celebrity beauty rankings from the popular press, and various industry publications.

Up to 15 weeks of revenue are analyzed for movies released in 2005. Boxofficemojo.com data are used to obtain weekly receipts. Movies with omitted budget information or limited releases, Japanese animation, IMAX films, and adult features are excluded. The resulting sample consists of 1539 observations pertaining to 140 movies. The average gross revenue is $55 million, and the average budget is $46 million, which is consistent with descriptive statistics presented in previous literature.

The Yahoo! Movies portal is used to track online WOM information for these 140 movies, because it is reasonable to believe that it mimics offline WOM dynamics. A total of 193,267 comments posted by viewers on Yahoo! Movies pertaining to the first 15 weeks of release (or fewer, if the movie was in release for a shorter time period) are used to provide estimates of volume as well as the valence of WOM.

5. Models

Separate models are estimated for the opening week and the subsequent weeks of the movies' theatrical runs. The study relies on the dynamic simultaneous equations method introduced into the movie domain by Elberse and Eliashberg (2003) and extended by Basuroy et al. (2006) to capture the endogenous relationships among star buzz, movie buzz, revenue, and the number of screens. Buzz around a film prior to its opening could potentially make consumers more interested in seeing the film. Exhibitors could increase (or decrease) the number of screens allocated to the film in response to this demand. The more screens the movie plays on, the more revenue it generates and the greater the number of consumers who see the film and are able talk about it with their peers. This, in turn, creates additional buzz and brings more people to the theaters. Subsequently, a film's popularity and slow revenue decay would push the distributors to keep the movie playing on a larger number of screens rather than to rapidly cut back to allow screen-allocation decisions to increase revenue once again.

Antecedents of star and movie buzz are included as independent variables in the STAR BUZZ and MOVIE BUZZ equations, whereas the revenue (distribution intensity) consequences are modeled in the REVENUE (SCREENS) equation. In addition to the variables explicitly outlined in the conceptual framework, the model also captures the impact of other traditional drivers such as seasonality, competition for screens from new movies, competition for revenue from similar movies, Motion Picture Association of America (MPAA) rating, other movie-related descriptors, and critical reviews. All variables, the rationale for their inclusion, and links to previously developed hypotheses are presented in Appendix A.

Opening Week_(for t = 1)
MOVIE BUZZ

$$MOVIE\_BUZZ_{it} = e^{\beta_0} \cdot (STAR\_BUZZ)_{it}^{\beta_2} \cdot e^{\beta_3 DMBi} \cdot e^{\varepsilon MBit} \quad (1)$$

STAR BUZZ

$$STAR\_BUZZ_{it} = e^{\alpha_0} \cdot (MOVIE\_BUZZ)_{it}^{\alpha_1} \cdot Z_{SBi}^{\alpha_2} \cdot e^{\alpha_3 DSBi} \cdot e^{\varepsilon SBit} \quad (2)$$

REVENUE

$$\begin{aligned} REVENUE_{it} = {} & e^{\gamma_0} \cdot (SCREENS)_{it}^{\gamma_1} \cdot (STAR\_BUZZ)_{i(t-1)}^{\gamma_2} \cdot \\ & (MOVIE\_BUZZ)_{i(t-1)}^{\gamma_3} \cdot X_{Rit}^{\gamma_4} \cdot Z_{Ri}^{\gamma_5} \cdot e^{\gamma_6 YRit} \cdot e^{\varepsilon Rit} \end{aligned} \quad (3)$$

SCREENS

$$\begin{aligned} SCREENS_{it} = {} & e^{\delta_0} \cdot (REVENUE^*)_{it}^{\delta_1} \cdot (STAR\_BUZZ)_{i(t-1)}^{\delta_2} \cdot \\ & (MOVIE\_BUZZ)_{i(t-1)}^{\delta_3} \cdot X_{Sit}^{\delta_4} \cdot Z_{Si}^{\delta_5} \cdot e^{\delta_6 DSi} e^{\delta_7 YSit} \cdot e^{\varepsilon Sit} \end{aligned} \quad (4)$$

The MOVIE BUZZ equation includes concurrent STAR $BUZZ_{it}$ and a vector of dummy control variables including BIG AD $BUDGET_i$, $SEQUEL_i$, and $FOREIGN_i$.

The STAR BUZZ equation includes concurrent MOVIE $BUZZ_{it}$. Z_{SBi} is a vector of time-invariant antecedents of star buzz such as STAR $BANKABILITY_i$ and STAR ACADEMY $AWARDS_i$.[2] D_{SBi} is a vector of dummy variables capturing the remaining antecedents such as STAR RECENCY OF $SUCCESS_i$ and STAR SEX $APPEAL_i$.

In the REVENUE equation, $SCREENS_{it}$ is the number of screens allocated by the distributors for movie *i* in week *t*. Also included are STAR $BUZZ_{i(t-1)}$ and MOVIE $BUZZ_{i(t-1)}$ corresponding to buzz generated during the week leading up to the opening. X_{Rit} is a vector of time-varying variables that includes the descriptors of the aforementioned movie buzz, namely PREV WEEK MOVIE BUZZ $VALENCE_{it}$ and PREV WEEK MOVIE BUZZ $INCONSISTENCY_{it}$. Also included in this vector are time-varying control variables such as $SEASONALITY_{it}$, COMPETITION REVENUE $RATINGS_{it}$, COMPETITION REVENUE $GENRE_{it}$, and PREV WEEK BOXOFFICE $RANK_{it}$. Z_{Ri} is the vector of time-invariant variables (e.g., CRITICAL $REVIEWS_i$) pertaining to the movies in question. The final term is the interaction between the logs of STAR BUZZ $_{i(t-1)}$ and MOVIE BUZZ $_{i(t-1)}$.

In the SCREENS equation for the opening week, we follow Basuroy et al. (2006) and use actual revenue for the first week. $REVENUES_{it}$ denotes the opening week revenue for movie *i* rather than consumer expectations from online simulation games such as Hollywood Stock Exchange (HSX), despite the availability of such data due to the potentially confounding effect on the analysis (e.g., Karniouchina (2011) empirically demonstrates that HSX users overprice movies with highly visible stars). Also included are STAR BUZZ $_{i(t-1)}$ and MOVIE BUZZ $_{i(t-1)}$. X_{Sit} is a vector of time-varying variables such as PREV WEEK MOVIE BUZZ $VALENCE_{it}$, COMPETITION SCREENS NEW $MOVIES_{it}$, and Z_{Si} controls for movie BUDGET, and D_{Si} is a vector of dummy variables that controls for screening intensity differences

[2] These factors do change over time, but they are assumed constant for the purposes of this investigation. Only pre-release critical acclaim and bankability are considered; focal movie is not included in the calculations.

related to genre and rating (R-RATING$_i$ and ANIMATION$_i$). The final term is the interaction between logs STAR BUZZ $_{i(t-1)}$ and MOVIE BUZZ $_{i(t-1)}$.

Later run
MOVIE BUZZ

$$MOVIE\_BUZZ_{it} = e^{\beta_0} \cdot (REVENUE)^{\beta_1}_{i(t-1)} \cdot (STAR\_BUZZ)^{\beta_2}_{it} \cdot X^{\beta_3}_{Bit} \cdot e^{\beta_4 DBit} \cdot e^{\varepsilon MBit} \tag{5}$$

STAR BUZZ

$$STAR\_BUZZ_{it} = e^{\alpha_0} \cdot (MOVIE\_BUZZ)^{\alpha_1}_{it} \cdot X^{\alpha_2}_{Bit} \cdot Z^{\alpha_3}_{Bi} \cdot e^{\alpha_4 DBit} \cdot e^{\varepsilon SBit} \tag{6}$$

REVENUE

$$REVENUE_{it} = e^{\gamma_0} \cdot (SCREENS)^{\gamma_1}_{it} \cdot (STAR\_BUZZ)^{\gamma_2}_{i(t-1)} \cdot (MOVIE\_BUZZ)^{\gamma_3}_{i(t-1)} \cdot X^{\gamma_4}_{Rit} \cdot Z^{\gamma_5}_{Ri} \cdot e^{\gamma_6 YRit} \cdot e^{\varepsilon Rit} \tag{7}$$

SCREENS

$$SCREENS_{it} = e^{\delta_0} \cdot (REVENUE^*)^{\delta_1}_{i(t-1)} \cdot (STAR\_BUZZ)^{\delta_2}_{i(t-1)} \cdot (MOVIE\_BUZZ)^{\delta_3}_{i(t-1)} \cdot X^{\delta_4}_{Sit} \cdot e^{\delta_5 DSi} e^{\delta_6 YSit} \cdot e^{\varepsilon Sit} \tag{8}$$

For the later run of the movie, the MOVIE_BUZZ equation is modified to include REVENUE $_{i(t-1)}$, PREV WEEK MOVIE BUZZ VALENCE, and PREV WEEK MOVIE BUZZ INCONSISTENCY. The STAR BUZZ equation now includes PREV WEEK MOVIE BUZZ VALENCE$_{it}$ (captured in X$_{Bit}$ as it is time-variant measure). The REVENUE equation now also includes the variables related to the user reaction to the movie, that is, PREV WEEK MOVIEBUZZ VALENCE$_{it}$ and PREV WEEK MOVIE BUZZ INCONSISTENCY$_{it}$. Also added is the PREV WEEK BOXOFFICE RANK$_{it}$ variable, which captures broad competitive dynamics that influence typical decay in revenue over time. The SCREENS equation is modified to include the revenue expectation $REVENUE_{it}{}^*$, which captures exhibitor expectations based on the exponential revenue decay between weeks (see Basuroy et al., 2006; Elberse & Eliashberg, 2003) and the viewer evaluation of feature quality (PREV WEEK MOVIE BUZZ VALENCE).

6. Estimation and model fit

6.1. Model fit

OLS, 2SLS, and 3SLS procedures are used to estimate log–log versions of opening week and later-run equations. The preferred estimation procedure in this setting is 3SLS, because OLS in inconsistent in the presence of endogeneity and 2SLS fails to capture contemporaneous cross-equation error correlation. Furthermore, the presence of lagged endogenous variables makes OLS biased. In 2SLS and 3SLS models, movie buzz, star buzz, revenue, and screens variables are treated as endogenous. The Hausman m-statistic (Wu, 1973) supports the use of the 3SLS approach for both the opening week and later-run samples used in this study. Results for the STAR BUZZ, MOVIES BUZZ, REVENUE, and SCREENS equations are presented in Table 2 through Table 6. All equations are identified, and the number of unique independent variables is equal to or exceeds the number of endogenous variables to be estimated (unique variables in each equation are highlighted).

6.2. Robustness checks

Robustness checks indicate that opening week vs. weeks 2 through 15 is an appropriate split. Estimating opening week equations over both week 1 and week 2 produces a decreased model fit compared to models that assume a 1-week opening period. However, later-run models for weeks 3+, weeks 4+, and so forth fit slightly better than the week 2+ models when it comes to the REVENUE and SCREENS equations and slightly worse when it comes to the two BUZZ equations. Because there are some interesting differences between the estimates obtained using these rival splits compared to the week 2+ models, their results are presented as well.

7. Findings

7.1. Antecedents of movie buzz

MOVIE BUZZ results (Table 1) support the existence of an endogenous relationship between star buzz and movie buzz during the opening week because the STAR BUZZ variable is significant ($p<0.01$). Buzz is stronger for movies that people like (H1 is supported); in other words, the valence of the previous week's user ratings is positively associated with the amount of movie buzz. H2 is supported as well: the inconsistency in user reviews is associated with higher buzz, which indicates that buzz is also propelled by people's desire to debate the quality of films. We also see that the previous week's box office receipts are associated with higher buzz. Several control variables are significant during the later run of the film, and increased movie buzz is also associated with movies that have big advertising budgets. Online buzz is lower for foreign features but higher for sequel films, which are more highly anticipated.

7.2. Antecedents of star buzz

The results (see Table 2) indicate that two of the four star-power factors were significantly related to the amount of star buzz during the opening week. SEX APPEAL was found to be significant (supporting H6). STAR ACADEMY AWARDS was found to be significant (supporting H5) but only at the $p<0.10$ significance level. Previous-week movie buzz was also highly significant, which supports the appropriateness of our conceptual framework. During the later run of a movie, star buzz is positively associated with several traditional star-power-related aspects, namely, coefficients for STAR BANKABILITY, STAR ACADEMY, and STAR SEX APPEAL. All of these are both positive and significant ($p<0.01$). Star buzz is also strongly associated with previous-week movie buzz. It is also associated with the valence of that buzz (for week 2+ equations only) at the $p<0.10$ significance level, which indicates that people are more likely to look up/talk about stars they saw in good movies. It is noteworthy that STAR SEX APPEAL is the only variable that is linked to star buzz at the 95% confidence level in both opening week and later-run models, which explains Hollywood's frequently criticized obsession with physical appearance.

7.3. Revenue consequences of buzz

As indicated in Table 3, pre-release movie and star buzz are associated with improved opening week box office performance supporting H7a and H9a. Additionally, the interaction between star and movie buzz during the opening week is positively associated with revenue, indicating that highly buzzed movies with highly visible stars enjoy significantly stronger openings and supporting H10a. Similar to previous findings, the number of screens has a positive association with box office performance. The later-run model also picks up significant effects of seasonality. Similar to opening week results and consistent with H7b, increased movie buzz is associated with higher revenue. However, contrary to H9b, during the later run of the movie, STAR BUZZ is associated with *lower* revenue. The STAR BUZZ variable has a significant ($p<0.10$) *negative* coefficient in week 2+ models, which becomes significant in the weeks 3+ and weeks 4+ REVENUE models. The interaction between the two types of

Table 1
3SLS results for MOVIE BUZZ.

	Opening week		*Later run*					
	Week 1		*Weeks 2+*		*Weeks 3+*		*Weeks 4+*	
Equation	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value
CONSTANT	− 0.010	0.94	− 1.824	0.00	− 1.579	0.00	− 1.456	0.00
STAR_BUZZ$_{it}$	0.825	0.00	0.17	0.00	0.192	0.00	0.194	0.00
BIG_AD_BUDGET$_i$[3]	0.035	0.63	0.169	0.00	0.193	0.00	0.218	0.00
PREV_WEEK_REVENUE$_{it}$	*NA*	*NA*	0.469	0.00	0.425	0.00	0.389	0.00
MOVIE_BUZZ_VALENCE$_{it}$	*NA*	*NA*	0.408	0.00	0.449	0.00	0.525	0.00
MOVIE_BUZZ_INCONSISTENCY$_{it}$	*NA*	*NA*	0.912	0.00	0.859	0.00	0.829	0.00
SEQUEL$_i$	0.035	0.62	0.1	0.00	0.091	0.00	0.091	0.00
FOREIGN$_i$	− 0.004	0.94	− 0.123	0.00	− 0.137	0.00	− 0.15	0.00
R^2	*0.470*		*0.7898*		*0.775*		*0.7478*	

[a]Unique variables in each equation are highlighted.

buzz is positive and significant in the weeks 3+ model ($p<0.10$) and significant in the weeks 4+ model ($p<0.05$), which indicates support for H10b.

Similar to Liu (2006), the results of this study show no evidence that the valence or consistency of WOM directly influences revenue. Therefore, H8 is not supported. Consistent with Godes and Mayzlin (2004), Dellarocas et al. (2007), and Dellarocas and Narayan (2006), this study finds an indirect relationship between movie buzz characteristics and revenue. According to the MOVIE BUZZ equation summarized earlier (Table 1), buzz characteristics (i.e., valence and consistency) affect the amount of buzz, which in turn influences the revenue.

Several control variables have been found to be significant as well. The presence of movies of the same genre and rating is associated with increased revenue. This surprising finding could be a result of studios' tendency to tailor movies to specific occasions. For instance, family movies come out at Christmas, horror flicks are released at Halloween, and animated films come out during school breaks. This creates situations in which similar flagship movies compete head-to-head. The variable COMPETITION REVENUE RATINGS is significant for the weeks 2+ model only (and marginally significant for the weeks 3+ model) while COMPETITION REVENUE GENRE is significant for all later-run data splits.

The model appropriately picks up the influence of critical reviews on movie performance. Previous studies have often included critical reviews in the equations for the opening week and omitted them thereafter under the assumption that their influence decays after the movie is released and its quality is revealed (Eliashberg & Shugan, 1997). The decay of the influence of critical reviews was a reasonable assumption in the past, when reviews appeared in newspaper columns prior to the release date. Currently, reviews are available online and consumers may consult them at any time after the movie is released. The significant coefficient associated with CRITICAL REVIEWS suggests that such an approach is indeed more appropriate.

7.4. The screening intensity consequences of buzz

The results (see Table 4) reveal strong support for H11a, H11b, and H11c during the opening week. Distributors lower the number of screens allocated for highly buzzed films (H11b) and films with visible celebrities (H11a) during the opening week. The interaction between the two types of buzz also has a marginally significant negative coefficient (H11c). It is possible that distributors employ this strategy to create greater hype around these films to increase their perceived desirability. This finding is also consistent with a belief that distributors expect viewers to go the extra mile to find an available screening location for a popular film. Another explanation is that the exhibitors prefer that their viewers see the movie later in its run (Morgan, 2009), because, although Hollywood is obsessed with opening week numbers as a measure of success, exhibitor fees are reduced during the later run. Therefore, it is profitable for exhibitors to lower the number of screens during the opening week and spread the demand, allocating more revenue to later periods. Because this can only be done for movies with a lot of buzz and interest around them, it is not surprising to find the significant negative coefficient corresponding to the MOVIE BUZZ variable.

Table 2
3SLS results for STAR BUZZ.

	Opening week		*Later run*					
	Week 1		*Weeks 2+*		*Weeks 3+*		*Weeks 4+*	
Equation	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value
CONSTANT	− 0.163	0.40	− 2.910	0.00	− 2.817	0.00	− 2.747	0.00
STAR_BANKABILITY$_i$[4]	0.037	0.45	0.090	0.01	0.116	0.00	0.145	0.00
STAR_RECENCY_OF_SUCCESS$_i$	0.020	0.68	0.031	0.39	0.030	0.41	0.026	0.51
STAR_ACADEMY AWARDS$_i$	0.050	0.06	0.065	0.00	0.062	0.00	0.067	0.00
STAR_SEX_APPEAL$_i$	0.360	0.00	0.492	0.00	0.483	0.00	0.489	0.00
MOVIE_BUZZ$_{it}$	1.098	0.00	0.420	0.00	0.416	0.00	0.385	0.00
MOVIE_BUZZ_VALENCE$_{it}$	*NA*	*NA*	0.219	0.08	0.192	0.13	0.207	0.12
R^2	*0.520*		*0.381*		*0.361*		*0.3368*	

[a]Unique variables in each equation are highlighted.

Table 3
3SLS results for REVENUE.

	Opening week		*Later run*					
	Week 1		*Weeks 2+*		*Weeks 3+*		*Weeks 4+*	
Equation	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value
CONSTANT	5.378	0.00	2.253	0.00	2.722	0.00	2.466	0.00
MOVIE_BUZZ $_{i(t-1)}$	0.231	0.03	0.240	0.01	0.328	0.00	0.437	0.00
PREV_WEEK_MOVIE_BUZZ_VALENCE$_{it}$	*NA*	*NA*	0.085	0.28	0.022	0.75	– 0.051	0.45
PREV_WEEK_MOVIE_BUZZ_INCONSISTENCY$_{it}$	*NA*	*NA*	– 0.028	0.76	– 0.008	0.92	– 0.035	0.63
STAR_BUZZ $_{i(t-1)}$	1.220	0.00	– 0.138	0.07	– 0.209	0.00	– 0.231	0.00
(MOVIE BUZZ $_{i(t-1)}$ x (STAR BUZZ $_{i(t-1)}$	0.370	<0.01	0.031	0.56	0.104	0.05	0.139	0.01
SCREENS$_{it}$	0.943	0.00	0.942	0.00	0.885	0.00	0.927	0.00
SEASONALITY$_{it}$[5]	0.032	0.68	0.222	0.00	0.179	0.00	0.16	0.00
COMPETITION_REVENUE_RATINGS$_{it}$	0.001	0.99	0.069	<0.05	0.050	0.08	0.041	0.12
COMPETITION_REVENUE_GENRE$_{it}$	– 0.024	0.77	0.157	0.00	0.149	0.00	0.155	0.00
CRITICAL_REVIEWS$_i$	0.167	0.34	0.693	0.00	0.621	0.00	0.589	0.00
PREV_WEEK_BOXOFFICE_RANK$_{it}$	*NA*	*NA*	0.183	0.52	0.425	0.11	0.31	0.24
R^2	*0.848*		*0.928*		*0.934*		*0.936*	

[a]Unique variables in each equation are highlighted.

It is interesting that distributor behavior mirrors that of audience behavior. During the later run of the film, distributors use a "push" strategy for most movies with buzzed stars (H11a) while also cutting back on their distribution of buzzed movies (H11b). This is especially true when this buzz is amplified by the celebrity buzz (H11c). It seems that distributors leave it to these movies to sell themselves and to consumers to go the extra mile to get to the available screening. It is worth noting that estimating a simple OLS regression with REVENUE as a dependent variable produces a positive and significant coefficient on the STAR BUZZ variable. If one considers the number of screens the movie is playing on, the results suggest that consumers are generally cautious of buzzed stars. Although the overall revenue associated with movies involving buzzed actors may be higher, taking revenue and screen number endogeneity into account allows us to see that this phenomenon occurs partially due to a distributor push strategy rather than consumer interest. In addition, movies with an R-RATING receive a more limited screening whereas ANIMATION features enjoy higher distribution.

7.5. Holdout sample validation

Model validation with a holdout sample is performed to ensure the robustness of the findings. The 140 films are randomly split into two subsets of 70, and the resulting opening week estimation and holdout samples have 70 observations each and the resulting later-run subsets differ by 4 observations due to the different run lengths of the films in the two subsets. Models are estimated on the subset containing more observations, and the estimates are used to predict the dependent variable for the smaller-holdout subset. Model estimates obtained from the calibration set are consistent with presented results obtained from the full sample. Correlations between the observed and predicted values in the validation set for the MOVIE BUZZ, STAR BUZZ, REVENUE, and SCREENS equations for the opening and later run were significant, with values of (0.57, 0.82), (0.53, 0.50), (0.83, 0.92), and (0.93, 0.96), respectively.

7.6. Quantifying the impact of star buzz

It is evident from the results that movie buzz is a major driving force behind box office success. However, the impact of star buzz is not as clear. On one hand, star buzz benefits the opening week performance. On the other hand, star buzz reduces performance during the later run of the film (in the weeks 2+ revenue model, it has a marginally significant negative coefficient, and starting with the weeks 3+ model, it has a highly significant negative impact). In addition, the interaction of star buzz with movie buzz is positive for

Table 4
3SLS results for SCREENS.

	Opening week		*Later run*					
	Week 1		*Weeks 2+*		*Weeks 3+*		*Weeks 4+*	
Equation	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value	Coeff.	*p*-value
CONSTANT	– 6.389	0.00	– 0.725	0.07	– 0.811	0.03	– 1.157	0.00
REVENUE*it	1.049	0.00	0.438	0.00	0.500	0.00	0.552	0.00
PREV_WEEK_MOVIE_BUZZ_VALENCE$_{it}$	*NA*	*NA*	0.143	0.02	0.194	0.00	0.218	0.00
STAR_BUZZ$_{i(t-1)}$	– 1.017	0.03	0.273	0.00	0.244	0.00	0.215	0.00
MOVIE_BUZZ$_{i(t-1)}$	– 0.589	0.02	– 0.193	0.04	– 0.195	0.04	– 0.227	0.03
(MOVIE BUZZ$_{i(t-1)}$ x (STAR BUZZ$_{i(t-1)}$	– 0.372	0.05	– 0.219	0.00	– 0.189	0.00	– 0.174	0.00
R-RATING$_i$[6]	0.028	0.65	– 0.142	0.00	– 0.143	0.00	– 0.131	0.00
ANIMATION$_i$	– 0.060	0.54	0.099	0.00	0.102	0.00	0.107	0.00
COMPETITION_SCREENS_NEW_MOVIES$_{it}$	– 0.024	0.70	0.031	0.12	– 0.009	0.64	– 0.009	0.60
BUDGET$_i$	0.079	0.30	0.103	0.00	0.072	0.02	0.076	0.01
R^2	*0.778*		*0.846*		*0.869*		*0.875*	

[a]Unique variables in each equation are highlighted.

the later run. Therefore, the overall impact of increased star buzz was evaluated by performing model simulations.

Traditionally, the impact of marketing actions is estimated by looking at elasticity of sales vis-à-vis marketing spending easily obtainable from log–log models (e.g., a 1% increase in advertising leads to β% increase in sales). However, this article deals with a more abstract star buzz factor as an independent variable, and a 1% increase in the buzz factor may not be a very useful metric from a managerial point of view. Therefore, to estimate the impact of increased star buzz, it is necessary to create a realistic scenario that mimics the impact of an effective PR campaign. To this end, model coefficients are used to compare the hypothetical performance of films employing stars who have average buzz factors with the performance of films with high-profile celebrities. To enhance the realism of the scenario, the buzz factor is allowed to vary by week, according to changes in the intensity of the campaign and consumer attention (e.g., variability in appearance intensity and venues and competing PR campaigns).

Separate equations are estimated for each week of the later run (e.g., models 5 through 8 were used to separately estimate equations for each week of the later run). The simulation relied on average estimates from the sample corresponding to the period being studied (i.e., separate estimates for each week) and positive star-buzz shocks randomly ranging from 1/2 to 1 standard deviation above the sub-sample mean corresponding to the appropriate period (the exact magnitude of a shock was drawn from a uniform distribution in that range). The positive impact of increased star buzz on an average movie's revenue is approximately $12 million during the opening week. Revenue is reduced by about $1 million during the later run, making the overall impact equal to approximately $11 million. Even for a movie with low movie buzz (1 standard deviation below the mean), an increase in star buzz results in roughly a $6 million increase in overall revenue. It is not clear if the reduced performance carries over to other markets (such as DVD sales). However, based on the box office data, it appears that box office performance is generally enhanced by increased star buzz.

8. Summary and future research directions

8.1. Summary of findings

The results indicate that stars have an impact on revenue, primarily due to their ability to generate buzz and drive audiences to the theaters during the opening week. In this sense, the findings of this study are similar to those of Ravid (1999), who suggests that stars set the floor for the revenue. However, this study shows that stars play a much more important role in enhancing box office revenue than suggested by previous studies and create additional buzz that has not been captured before. The results indicate that star buzz has the potential to enhance opening week box office receipts both directly and by contributing to the overall movie anticipation (or pre-release movie buzz). However, star buzz also has a negative impact on revenue during subsequent weeks. Despite this effect, model simulations showed that the overall impact of star buzz is positive because the initial revenue boost outweighs the later decline. Overall, the results suggest that star and movie buzz and their interaction are the leading explanatory factors of both box office revenue and screen allocation. The summary of the findings with respect to the original hypotheses is presented in Table 5.

This study also helps shed some light on the role of more traditional measures of star power such as bankability, sex appeal, and *Academy Awards*. Sex appeal increases star buzz during both the opening and later-run periods of the film. *Academy Award* nominations are significant at the $p<0.10$ level in increasing star buzz during the opening week and at the $p<0.05$ level during the later run of the

Table 5
Summary of results.

Support / Hypothesis	Opening week	Later run
H1: The valence of movie buzz is positively associated with the volume of movie buzz.	NA	Supported ($p<0.01$)
H2: Movie buzz inconsistency is positively associated with the volume of movie buzz.	NA	Supported ($p<0.01$)
H3: Bankability is positively associated with star buzz.	Not Supported	Supported ($p<0.05$)
H4: Recency of success is positively associated with star buzz.	Not Supported	Not Supported
H5: Academy Awards are positively associated with star buzz.	Supported ($p<0.10$)	Supported ($p<0.01$)
H6: Sex appeal is positively associated with star buzz.	Supported ($p<0.01$)	Supported ($p<0.01$)
H7a: Pre-release movie buzz is positively associated with opening week box office revenue.	Supported ($p<0.05$)	NA
H7b: Post-release movie buzz is positively associated with later-run box office revenue.	NA	Supported ($p<0.01$)
H8: The valence of movie buzz is positively associated with box office revenue.	NA	Not Supported: only an indirect relationship is found.
H9a: Pre-release star buzz is positively associated with opening week box office revenue.	Supported ($p<0.01$)	NA
H9b: Post-release star buzz is positively associated with later-run box office revenue.	NA	Not Supported: the opposite relationship is found.
H9c: Star buzz is more important for boosting opening week than later-run revenue.	Supported: there is a strong positive impact of Star Buzz on opening week revenue, but the effect is negative during the later run.	
H10a: Interaction between star buzz and movie buzz is positively associated with opening box office revenue.	Supported ($p<0.01$)	NA
H10b: Interaction between star buzz and movie buzz is positively associated with later-run box office revenue.	NA	Partially Supported: significant for weeks 4+ ($p<0.05$)
H11a: Star buzz influences exhibitor decisions.	Supported ($p<0.05$), distributors allocate fewer screens to movies with high star buzz.	Supported ($p<0.05$), distributors allocate more screens to movies with high star buzz.
H11b: Movie buzz influences exhibitor decisions.	Supported ($p<0.05$), distributors allocate fewer screens to movies with high buzz around them.	Supported ($p<0.05$), distributors allocate fewer screens to movies with high buzz around them.
H11c: Interaction between star and movie buzz influences exhibitor decisions.	Supported ($p=0.05$).	Supported ($p<0.05$).

film. Long-term bankability (which reflects the level and consistency of past performances) is also positively associated with star buzz during the later run of the movie.

In addition to enriching the understanding of the role of star power in driving box office revenue, this is the first study to explicitly model the endogenous nature of star and movie buzz and their impact on box office revenue and distribution decisions. Hennig-Thurau, Houston, and Sridhar (2006) used structural equation modeling (SEM) to demonstrate that studio actions primarily impact early box office receipts, whereas feature quality has a short- and long-term influence on box office performance. Similarly, this study finds that star buzz is critical during the opening week and movie buzz drives sales throughout the entire theatrical release of the movie. Model simulations show that a self-regulating mechanism used by consumers (which punishes bad pictures for using buzz campaigns) provides a partial safeguard against consumer exploitation. However, the initial sales boost is greater than the future penalty. Future studies should examine whether there is a carryover effect to future projects of highly buzzed stars whose projects fall short of consumer expectations.

8.2. Implications

Overall, star buzz, as measured by IMDB searches, was found to be a very informative gauge of star power. Therefore, this research sheds light on the long-standing debate regarding the role of stars in the success of motion pictures. This study also suggests that simply entering a variable related to one of the traditional dimensions of star power (e.g., *Academy Awards* or bankability) in motion picture research is not sufficient. One must concentrate on a star's ability to grab consumer attention. This attention-grabbing mechanism allows stars to boost opening revenue significantly. Therefore, star buzz needs to be considered in future studies. Although previous research questioned the exorbitant fees demanded by high-profile actors, this research clearly demonstrates the positive impact of actor visibility on overall box office performance.

According to the simulation, under realistic conditions increased star buzz can be beneficial to the underlying projects. Therefore, production studios and distributors should take steps to enhance stars' public profiles prior to movie releases. Furthermore, studios should work with fan sites to spread the word about upcoming projects. One option for promoting fan activity would be to provide exclusive sneak peak opportunities. In addition to scheduling high-profile PR appearances, studios may also consider creating interactive and viral web content, because this research demonstrates the high value of cyber buzz.

The study also demonstrates the importance of pre-release movie buzz, which has been overlooked by previous research, and presents a new framework for incorporating star and movie buzz into motion picture performance modeling. Therefore, pre-release cyber buzz should be considered when building forecasts for other products outside of the movie industry. The results also clearly indicate that movie buzz is critical in boosting box office revenue beyond the opening week. Furthermore, the valence of buzz and its consistency are both related to the amount of buzz generated. People are more likely to talk about "better" movies and movies of debatable quality. Therefore, providing additional forums for information exchange, encouraging the debate about the movie quality, and investing in buzz-creating Internet activities are critical. Providing independent parties with exclusive sneak previews, insider photos, and other "juicy" information prior to the movie's release while also encouraging various types of games – e.g., those tied into social networking sites – can be very beneficial in promoting cyber buzz.

8.3. Limitations

Several ordinal measures such as search ranks are assumed to represent the actual counts of consumer search instances; even though the data to verify such an assumption are not available (IMDB provides the ranks, but not the number of hits). Although these assumptions are somewhat limiting and further research is needed to verify their appropriateness, industry ranks used in previous studies have received similar treatment, and thus have similar limitations. This study did not incorporate any variables related to the valence of star buzz. The introduction of such measures could further enrich our understanding of the buzz marketing phenomenon. However, large amounts of qualitative data and expert-based panel approaches must be used when separating star-related buzz into positives and negatives. The search-based data used in this study do not allow for this type of investigation. Another limitation of search-based data is that they do not allow the differentiation between marketer-initiated and organic types of buzz.

8.4. Research opportunities

Close examination of celebrity news archives indicates that celebrity news frequently coincides with movie premiers, which suggests that increased exposure is often deliberate rather than a mere spillover of celebrities' full, over-the-top lifestyles. For instance, Angelina Jolie had an "embarrassing" wardrobe malfunction at the premier of the movie *Alexander*. If deliberate, this publicity should be considered covert marketing because consumers may be unaware of the fact that they are being manipulated. Future studies should explore the variations of buzz in anticipation of movie releases to establish what proportion of buzz is artificially created in direct response to the upcoming movie. In this respect, it would be interesting to use the analytical approaches to events that are common in the finance literature and to look for abnormalities related to star buzz factors in response to movie openings and other events. It may also be beneficial to look for a clustering of artificial buzz-creating activities prior to movie openings.

The notion of buzz is relevant to many motion picture-related literature streams. For instance, the inclusion of star and movie buzz can improve demand-forecasting and movie-scheduling models (Eliashberg et al., 2009). It also can enrich our understanding of rent vs. buy decisions in the motion picture domain (Knox & Eliashberg, 2009). People may be willing to view a well-hyped movie, but they may be reluctant to spend the extra money to actually own it.

This study underscores the importance of using online search data, which are not limited to the motion picture industry. Search intelligence is quickly becoming a dominant area in marketing. *Hitwise.com*, a company that provides businesses with advanced analyses of search terms and rankings of hits on various web pages, has an extensive list of over 1,200 diverse clients including *Virgin*, *Panasonic*, *Heinz*, *Honda*, and *MTV*. Other sites affiliated with major search engines, such as buzz.yahoo.com or trends.google.com, provide analyses of emerging trends. For example, buzz.yahoo.com provides search ratings and dynamics pertaining to competing recording artists, video games, and other domains. Search data are increasingly used to evaluate the effectiveness of marketing and promotional campaigns as well as branding strategies. Due to the richness and timeliness of search-based data, there is potential for academics in the field of marketing to use such data in their research.

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Appendix A. Description of variables

Equation	Variable	Description and operationalization	Rationale for inclusion	Related hypothesis/modeling consideration/control variable
MOVIE BUZZ	MOVIE BUZZ	Variable indicates the amount of buzz that surrounds a movie and measures the intensity of consumer-driven information flow. The opening week equation uses the intensity of Internet searches. After the opening week, the intensity of online WOM is used: following Liu (2006), we use a count of user reviews posted on movies.yahoo.com.	DV	
	STAR BUZZ	Variable that captures the top Internet search rank of any star in the movie cast each week; it is a measure of public interest in the star. StarMeter rank reflects the intensity of Internet searches within Imdb.com portal. Reverse-coded for ease of interpretation.	This variable is included to capture the endogenous relationship between star buzz and movie buzz.	Modeling endogeneity
	BIG AD BUDGET	Dummy variable that takes on a value of "1" if the movie was mentioned in the *Hollywood Reporter*'s publication on the top advertisement spenders in 2005 and "0" otherwise.	Distributors may influence the amount of movie buzz through advertising.	Control
	PREV WEEK REVENUE	Weekly revenue for the week prior to the one in question (obtained from boxofficemojo.com).	The number of people who saw the movie during the previous week should have a positive association with movie buzz.	Modeling endogeneity
	MOVIE BUZZ VALENCE	Average user score for a given movie calculated based on the reviews posted during the week prior to the one in question converted to a numerical scale.	People could be more likely to talk about the movies they liked (Liu, 2006).	H1
	MOVIE BUZZ INCONSISTENCY	Standard deviation of user review scores described above if more than one score is present and 0 otherwise (it is assumed that there is no debate under this condition). Previous studies separated WOM into positive or negative, but this study assumes that the valence of buzz is equal to the scores consumers assign to the movie. This approach relies on consumer perception rather than researcher judgment and avoids collinearity between positive and negative WOM, which prevents the estimation of the models when both types of WOM are present.	Movie buzz could be higher if there is a debate regarding movie quality.	H2
	SEQUEL	Dummy variable indicating whether the film is a sequel.	Movie buzz could be stronger for sequels because they are generally made when the original movies resonate with the audiences; thus, the excitement and anticipation could be stronger compared to non-sequel films.	Control
	FOREIGN	Dummy variable indicating whether the film was produced outside the United States.	Buzz could be lower for foreign movies.	Control
STAR BUZZ	STAR BUZZ	See description in the Movie Buzz equation above.	DV	
	STAR BANKABILITY	Combines the rank of an actor based on lifetime per movie average and the rank based on the total number of movies in the casting history. Reverse-coded for ease of interpretation. The sum of the two ranks was used to rank-order all of the stars on boxofficemojo.com's People Index list supplemented with several omitted actors relevant to this study. The information on lifetime box office for these omitted actors was collected through Imdb.com. The top five bankable stars were Tom Hanks, Eddie Murphy, Harrison Ford, Tom Cruise, and Robin Williams.	Variable measures the star's stamina and prominence and avoids rewarding actors with too many episodic roles. Longevity and prominence could make consumers more likely to talk about stars.	H3
	STAR RECENCY OF SUCCESS	A dichotomous variable that indicates whether any member of the cast had a blockbuster hit (i.e., a movie grossing over $100 million) within the last three years. This variable is similar to the NEXT variable used by Ravid (1999) (which only captured the previous year) but allows for actor participation in independent films between "big" movies.	Consumers may be more likely to talk about stars of recent blockbusters.	H4
	STAR ACADEMY AWARDS	Represents the number of Academy Award winners and nominees in the cast (limited to the top three actors and excludes cameo appearances).	Industry recognition could fuel consumer interest and information exchange.	H5
	STAR SEX APPEAL	A dichotomous variable that represents whether any of the cast members were featured on various Hollywood beauty lists. Among the lists are E! "101 Celebrity Hottest Bodies," *People*'s "Most Beautiful People List," and *Teen People*'s "The 25 Hottest Stars under 25."	Consumers could be more likely to discuss attractive celebrities.	H6
	MOVIE BUZZ	See description in the Star Buzz equation above.	This variable is included in this equation to capture the endogenous relationship between star buzz and movie buzz.	Modeling endogeneity
	MOVIE BUZZ VALENCE	Average user score for a given movie calculated based on the reviews posted to movies.yahoo.com during a week prior to the one in question (stars are converted to a numerical scale).	People could be more likely to talk about stars who appear in movies they liked.	Modeling endogeneity
REVENUE	REVENUE	Weekly revenue figures for up to 15 weeks of theatrical release. Data were obtained from boxofficemojo.com.	DV	
	MOVIE BUZZ $_{t-1}$	MOVIE BUZZ during the week leading up to the one in question.		H7a and H7b
	PREV WEEK MOVIE BUZZ VALENCE	See MOVIE BUZZ VALENCE above. Same measure for the week leading up to the one in question.	People could be more likely to talk about good (or bad) films.	H8

Appendix A (*continued*)

Equation	Variable	Description and operationalization	Rationale for inclusion	Related hypothesis/modeling consideration/control variable
	PREV WEEK MOVIE BUZZ INCONSISTENCY	See MOVIE BUZZ INCONSISTENCY above. Same measure for the week leading up to the one in question.	People could be more reluctant to see the film if there is no agreement regarding its quality. Alternatively, if there is a debate regarding quality, people could be more likely to go and "see for themselves."	Control
	STAR BUZZ $_{t-1}$	Star buzz related to the week leading up to the one in question.		H9a, H9b, H9c
	SCREENS	Weekly screen count for up to 15 weeks of data.	The inclusion of this variable captures the supply and demand dynamics in theatrical distribution (Elberse & Eliashberg, 2003).	Modeling endogeneity
	SEASONALITY	Measures of underlying seasonality changes (based on Einav, 2007). Data provided courtesy of Professor Einav.	Captures seasonal fluctuations not related to movie quality.	Control
	COMPETITION REVENUE RATINGS	Number of movies in the top 20 during the previous week that have been assigned the same MPAA rating as the movie in question.	Controls for the impact on revenue associated with concurrent releases of similar films.	Control
	COMPETITION REVENUE GENRE	Number of movies in the top 20 during the previous week that have the same genre as the movie in question.	Controls for the impact on revenue associated with concurrent releases of similar films.	Control
	CRITICAL REVIEWS	Critical reviews were obtained from movies.yahoo.com, which compiles the reviews from various sources, such as *Chicago Tribune, Boston Globe, Entertainment Weekly, New York Times, San Francisco Chronicle,* and *New York Post,* and were converted to a numerical scale.	Controls for the impact of critical reviews on box office performance. See Basuroy et al. (2003) for more details.	Control
	PREV WEEK BOX OFFICE RANK	Box office standing from the previous week obtained from weekly listings on boxofficemojo.com and reverse-coded for ease of interpretation.	Captures broad competitive dynamics.	Control
	(MOVIE BUZZ $_{t-1}$) x (STAR BUZZ $_{t-1}$)	Interaction between logs of previous-week star and movie buzz or *log* (MOVIE BUZZ$_{t-1}$) × *log* (STAR BUZZ $_{t-1}$).	Captures moderating relationship.	H10a and H10b
SCREENS	SCREENS	Weekly screen count for up to 15 weeks of data.	DV	
	REVENUE*	Revenue expectation based on revenue history obtained from boxofficemojo.com, calculated as described in Elberse and Eliashberg (2003).	Captures supply and demand dynamics of theatrical distribution	Modeling endogeneity
	PREV WEEK MOVIE BUZZ VALENCE	See above.	Included to measure the impact of buzz on distribution.	H11b
	STAR BUZZ $_{t-1}$	Star buzz during the week leading up to the one in question.	Included to measure the impact of buzz on distribution.	H11a
	MOVIE BUZZ $_{t-1}$	Movie buzz during the week leading up to the one in question.	Included to measure the impact of buzz on distribution.	H11b
	(MOVIE BUZZ $_{t-1}$) x (STAR BUZZ $_{t-1}$)	Interaction between the logs of star and movie buzz or *log* (MOVIE BUZZ $_{t-1}$) × *log* (STAR BUZZ $_{t-1}$)	Included to measure the impact of buzz on distribution.	H11c
	R-RATING	Dummy variable that takes on a value of "1" when a movie is listed as having an "R" MPAA rating according to imdb.com.	R-rated films have a more limited distribution.	Control
	ANIMATION	Dummy variable that takes on a value of "1" when a movie is listed as an animation film on imdb.com.	Animated movies are generally believed to have a broader distribution.	Control
	COMPETITION SCREENS NEW MOVIES	Combined budget of new releases during the week in question. Information obtained from box officemojo.com and supplemented by imdb.com and trade publications.	New movies compete for screens.	Control
	BUDGET	Movie budget figures were obtained from various sources including imdb.com, boxofficemojo.com, and trade publications.	Traditional measure included in screens equations.	Control

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